BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 2: Validation, Testing and Debugging

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does selection do?

2. What is NOT true?
   A) true
   B) 0.5
   C) false
   D) An empty string

3. "A Boolean variable represents any unrestricted paragraph of text." Correct this. Explain why.

4. A program joins strings and takes part of an eight-character word. Correct both claims. Explain why. (a) Joining strings adds their numbers. (b) substring(1,3) always means positions 1 to 3 in every language.

Today's objectives

Explain the core ideas of paper 2: validation, testing and debugging using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Validation

Test data against defined rules and loop back after rejection if further input is required.

Testing

Include normal, boundary, invalid and erroneous inputs.

Logic errors

A syntactically valid program may still use incorrect conditions or updates.

Maintainability

Simple named variables, comments and subprograms support future debugging.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Validation
Test data against defined rules and loop back after rejection if further input is required.

Testing
Include normal, boundary, invalid and erroneous inputs.

Logic errors
A syntactically valid program may still use incorrect conditions or updates.

Maintainability
Simple named variables, comments and subprograms support future debugging.

I Do - Worked model

A test plan is needed for a quantity 1-20 inclusive. Produce four categories.

- Normal quantity=10.
- Boundary quantities=1 and 20.
- Invalid numeric quantities 0 and 21.
- Erroneous input 'many' should be handled safely.

We Do - Guided practice

1. Fix a quantity check using quantity>=1 OR quantity<=20.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Fix the range check for an image width of zero pixels.

```
width = 0
if width >= 1 OR width <= 1920 then
print("Valid")
else
print("Invalid")
endif
```

You Do - Independent exam questions

Q2. Give one invalid and one erroneous input for age 5-16. [2 marks]

Q3. Explain the difference between validation and authentication. [2 marks]

Q4. Describe how a trace table helps locate a logic error. [3 marks]

Exit Ticket - True or False?

Boundary testing includes the smallest and largest valid values.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It chooses a branch using a condition. Selection can be nested.

2. C - false

3. True or false.

4. (a) Joining 'AB' and '12' gives 'AB12', with no added space. (b) For 'COMPUTER', substring(1,3) in OCR ERL gives 'OMP': indexes start at 0 and 3 is the length. Follow each language's rules.

I Do - Explain the model

Normal quantity=10.

Boundary quantities=1 and 20.

Invalid numeric quantities 0 and 21.

Erroneous input 'many' should be handled safely.

We Do - Model answers

1. Accept 1 and 20, reject 0 and 21.

Code First - Answer

```
width = 0
if width >= 1 AND width <= 1920 then
print("Valid")
else
print("Invalid")
endif
```

Expected output: Invalid

You Do - Self-mark

Q2: Invalid 20 (integer out of range); erroneous 'cat' (non-numeric text).

Q3: Validation checks input constraints; authentication checks claimed identity.

Q4: It records successive states and shows the first variable/branch differing from expected behaviour.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.